Domain and Range

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Domain from equations**

State the domain of each function:

a $f(x) = \frac{1}{x-4}$ All real x with $x \neq 4$.

b $g(x) = \sqrt{2x-6}$ $2x - 6 \geq 0$, so $x \geq 3$.

c $h(x) = \frac{x+1}{x^2-9}$ All real x with $x \neq 3$ and $x \neq -3$.

d $k(x) = \sqrt[3]{x}$ All real numbers.

(a) All real x with $x \neq 4$.; (b) $2x - 6 \geq 0$, so $x \geq 3$.; (c) All real x with $x \neq 3$ and $x \neq -3$.; (d) All real numbers.

2 **[Foundational]** • **Target: Domain from equations**

Find the domain of $f(x) = \frac{\sqrt{5-x}}{x-1}$. Write your answer in interval notation.

We need $5 - x \geq 0$, so $x \leq 5$, and $x \neq 1$. Domain: $(-\infty, 1) \cup (1, 5]$.

3 **[Foundational]** • **Target: Domain and range from graphs**

The graph of $y = x^2 - 2$ is shown. State the domain and range of the function.

Domain: all real numbers. Range: $y \geq -2$.

4 **[Foundational]** • **Target: Domain and range from graphs**

The graph of $y = \sqrt{x+3}$ is shown. State the domain and range of the function.

Domain: $x \geq -3$. Range: $y \geq 0$.

5 **[Foundational]** • **Target: Finding the range algebraically**

By completing the square, find the range of $f(x) = x^2 + 4x + 1$.

$f(x) = (x+2)^2 - 3$. Since $(x+2)^2 \geq 0$, the range is $y \geq -3$.

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6 [Foundational] • Target: Finding the range algebraically

Consider $f(x) = -2(x-1)^2 + 5$.

a State the coordinates of the vertex. Vertex: $(1, 5)$.

b Does the parabola open upward or downward? Downward, since the leading coefficient $-2 < 0$.

c State the range of f. $y \le 5$.

(a) Vertex: $(1, 5)$.; (b) Downward, since the leading coefficient $-2 < 0$.; (c) $y \le 5$.

7 [Foundational] • Target: Finding the range algebraically

Find the range of $g(x) = \sqrt{9 - x^2}$ (the upper half of a circle).

The domain is $[-3, 3]$; g increases from $g(\pm 3) = 0$ to a maximum $g(0) = 3$. Range: $0 \le y \le 3$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Domain and range in applied contexts

A rectangular garden has perimeter 40 m. If x is the width, the length is $20 - x$, and the area is $A(x) = x(20 - x)$.

a State the domain of A in this context (both dimensions must be positive). $0 < x < 20$.

b Find $A(6)$ and $A(15)$. $A(6) = 6(14) = 84$ m^2; $A(15) = 15(5) = 75$ m^2.

c State the range of A over this domain, to the nearest whole number. $A(x) = -(x-10)^2 + 100$ has maximum 100 at $x = 10$ and approaches 0 at the endpoints: range $0 < y \le 100$.

(a) $0 < x < 20$.; (b) $A(6) = 6(14) = 84$ m^2; $A(15) = 15(5) = 75$ m^2.; (c) $A(x) = -(x-10)^2 + 100$ has maximum 100 at $x = 10$ and approaches 0 at the endpoints: range $0 < y \le 100$.

9 [Intermediate] • Target: Domain and range in applied contexts

An open-top box is formed from a 12 cm square by cutting squares of side x from each corner, giving volume $V(x) = x(12 - 2x)^2$.

a State the domain of V in this physical context. $0 < x < 6$.

b Find $V(1)$ and $V(4)$. $V(1) = 1(10)^2 = 100$ cm^3; $V(4) = 4(4)^2 = 64$ cm^3.

(a) $0 < x < 6$.; (b) $V(1) = 1(10)^2 = 100$ cm^3; $V(4) = 4(4)^2 = 64$ cm^3.

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10 [Intermediate] • Target: Domain of composite and combined functions

Let $f(x) = \sqrt{x-2}$ and $g(x) = \frac{1}{x}$. Find the domain of:

a $(f+g)(x)$ Need $x \geq 2$ and $x \neq 0$: domain $x \geq 2$. **b** $(f \circ g)(x)$ $f(g(x)) = \sqrt{\frac{1}{x} - 2}$, needs $\frac{1}{x} \geq 2$ and $x \neq 0$: domain $0 < x \leq \frac{1}{2}$.

(a) Need $x \geq 2$ and $x \neq 0$: domain $x \geq 2$.; (b) $f(g(x)) = \sqrt{\frac{1}{x} - 2}$, needs $\frac{1}{x} \geq 2$ and $x \neq 0$: domain $0 < x \leq \frac{1}{2}$.

11 [Intermediate] • Target: Domain of composite and combined functions

Find the domain of $f(x) = \frac{1}{\sqrt{x^2-4}}$, writing your answer in interval notation.

Need $x^2 - 4 > 0$ (strictly, since it's in a denominator under a root): $x < -2$ or $x > 2$. Domain: $(-\infty, -2) \cup (2, \infty)$.

12 [Intermediate] • Target: Domain from rational functions with factoring

Find the domain of $f(x) = \frac{x^2-4}{x^2-x-6}$, factoring the denominator first.

$x^2 - x - 6 = (x-3)(x+2)$. Domain: all real x with $x \neq 3$ and $x \neq -2$.

13 [Intermediate] • Target: Domain from rational functions with factoring

Find the domain of $g(x) = \frac{\sqrt{x+4}}{x-1}$, combining a root restriction with a denominator restriction.

Need $x + 4 \geq 0 \Rightarrow x \geq -4$, and $x \neq 1$. Domain: $[-4, 1) \cup (1, \infty)$.

14 [Intermediate] • Target: Range using transformations

The range of $f(x) = x^2$ is $y \geq 0$. State the range of $g(x) = f(x) - 5 = x^2 - 5$ and of $h(x) = -f(x) = -x^2$, using transformation reasoning.

$g(x) = x^2 - 5$ shifts the range down 5: range $y \geq -5$. $h(x) = -x^2$ reflects the range over $y = 0$: range $y \leq 0$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Range using transformations • Cross-Lesson: 5.03 Parametric Equations

The range of $f(x) = \sqrt{x}$ is $y \geq 0$. State the range of $g(x) = 3\sqrt{x} + 2$.

Multiplying by 3 scales the range but preserves $y \geq 0$; adding 2 shifts up: range $y \geq 2$.

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16 [Challenging] • Target: Domain and range of piecewise functions • Cross-Lesson: 6.01 Polar Coordinates and Conversions

A function is defined by $f(x) = x^2$ for $x < 0$ and $f(x) = 2x + 1$ for $x \geq 0$. State the domain and range.

Domain: all real numbers (every x has a defined piece). For $x < 0$: $f(x) > 0$ (excluding 0 itself, approaching it). For $x \geq 0$: $f(x) \geq 1$. Combined range: $y > 0$.

17 [Challenging] • Target: Domain of composite and combined functions • Cross-Lesson: 6.02 Polar Equations and Graphs

Find all vertical, horizontal, and slant asymptotes and any removable discontinuities (holes) of the rational function $f(x) = \frac{3x^2+1}{x^2-4}$. Describe its end behavior using limits.

Step 1: Domain and vertical asymptotes: The denominator factors as $x^2 - 4 = (x-2)(x+2)$. The numerator $3x^2 + 1 > 0$ has no real zeros. Thus, there are no common factors and no removable holes. The vertical asymptotes are $x = 2$ and $x = -2$. Step 2: Horizontal asymptote and end behavior: Numerator and denominator have equal degree $n = d = 2$: $\lim_{x \to \pm\infty} f(x) = \lim_{x \to \pm\infty} \frac{3 + 1/x^2}{1 - 4/x^2} = \frac{3}{1} = 3.$ The horizontal asymptote is $y = 3$. Notice tha.

18 [Challenging] • Target: Domain from rational functions with factoring • Cross-Lesson: 7.01 Exponential Functions and Their Graphs

Decompose the rational expression $R(x) = \frac{7x-1}{x^2-x-6}$ into partial fractions using the Heaviside cover-up method.

Step 1: Factor denominator: $x^2 - x - 6 = (x-3)(x+2)$. Set up the decomposition: $\frac{7x-1}{(x-3)(x+2)} = \frac{A}{x-3} + \frac{B}{x+2}$. Step 2: Apply Heaviside cover-up method:For A (evaluate at $x = 3$): $A = \frac{7x-1}{x+2}\Big|_{x=3} = \frac{7(3)-1}{3+2} = \frac{20}{5} = 4.$ For B (evaluate at $x = -2$): $B = \frac{7x-1}{x-3}\Big|_{x=-2} = \frac{7(-2)-1}{-2-3} = \frac{-15}{-5} = 3.$ The partial fraction decomposition is $\frac{4}{x-3} + \frac{3}{x+2}$.

19 [Challenging] • Target: Range using transformations • Cross-Lesson: 7.02 Logarithms and Their Properties

What is the domain and range of $y = \tan^{-1}(x)$?

$\tan^{-1}(x)$ has domain $(-\infty, \infty)$ (restricted input) and range $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ (restricted principal output). Final Answer: Domain $(-\infty, \infty)$, range $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

20 [Challenging] • Target: Domain and range of piecewise functions • Cross-Lesson: 7.03 Exponential and Logarithmic Equations

In ΔABC, $a = 7$, $b = 9$, and $c = 10$. Find the measure of the smallest angle to the nearest tenth of a degree, and compute the area of ΔABC using Heron's formula.

Step 1: Identify smallest angle: The smallest angle is A, opposite the shortest side $a = 7$. Step 2: Law of Cosines for angle A: $\cos A = \frac{b^2+c^2-a^2}{2bc} = \frac{9^2+10^2-7^2}{2(9)(10)} = \frac{81+100-49}{180} = \frac{132}{180} = \frac{11}{15} \approx 0.7333.$ $A = \arccos(11/15) \approx 42.8^\circ.$ Step 3: Heron's formula for area: Semi-perimeter $s = \frac{7+9+10}{2} = 13.$.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Finding Domains of Rational and Radical Functions

Find the domain of $f(x) = \frac{x-3}{x^2-16}$.

a. $(-\infty, -4) \cup (-4, 4) \cup (4, \infty)$ ✓

b. $(-\infty, 4) \cup (4, \infty)$

c. $[-4, 4]$

d. All real numbers except $x = 3$

Denominator cannot be zero: $x^2 - 16 \neq 0 \Rightarrow x \neq \pm 4$. Domain: all reals except $x = -4$ and $x = 4$.

22 [Foundational] • Target: Finding Domains of Rational and Radical Functions

Find the domain of $g(x) = \sqrt{2x-8}$.

a. $[4, \infty)$ ✓

b. $(4, \infty)$

c. $[8, \infty)$

d. $(-\infty, 4]$

The radicand must be non-negative: $2x - 8 \geq 0 \Rightarrow 2x \geq 8 \Rightarrow x \geq 4$, which is $[4, \infty)$.

23 [Foundational] • Target: Finding Domains of Rational and Radical Functions

Find the domain of $h(x) = \frac{1}{\sqrt{9-x^2}}$.

a. $(-3, 3)$ ✓

b. $[-3, 3]$

c. $(-\infty, -3) \cup (3, \infty)$

d. $[0, 3)$

Radicand in denominator must be strictly positive: $9 - x^2 > 0 \Rightarrow x^2 < 9 \Rightarrow -3 < x < 3$.

24 [Foundational] • Target: Finding Ranges of Algebraic Functions

Find the range of $f(x) = x^2 - 4x + 7$.

a. $[3, \infty)$ ✓

b. $[7, \infty)$

c. $(-\infty, 3]$

d. $[0, \infty)$

Complete the square: $f(x) = (x-2)^2 + 3$. Since $(x-2)^2 \geq 0$, the minimum value is 3. Range: $[3, \infty)$.

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25 [Foundational] • Target: Finding Ranges of Algebraic Functions

Find the range of $g(x) = -|x + 2| + 5$.

a. $(-\infty, 5]$ ✓

b. $[5, \infty)$

c. $(-\infty, -2]$

d. $[-5, 5]$

Since $|x + 2| \geq 0$, $-|x + 2| \leq 0$. Adding 5 gives $g(x) \leq 5$. Range: $(-\infty, 5]$.

26 [Foundational] • Target: Finding Ranges of Algebraic Functions

Find the range of $h(x) = \sqrt{25 - x^2}$.

a. $[0, 5]$ ✓

b. $[-5, 5]$

c. $[0, 25]$

d. $[5, \infty)$

Domain is $[-5, 5]$. Min occurs at $x = \pm 5$ ($h = 0$). Max occurs at $x = 0$ ($h = 5$). Range is $[0, 5]$.

27 [Foundational] • Target: Piecewise Functions

Evaluate $f(-2)$ for $f(x)$ defined as $2x + 1$ for $x < 0$, and $x^2 - 3$ for $x \geq 0$.

a. -3 ✓

b. 1

c. -2

d. 7

Since $-2 < 0$, use the top rule: $f(-2) = 2(-2) + 1 = -4 + 1 = -3$.

28 [Foundational] • Target: Piecewise Functions

Find the domain of $f(x)$ defined as $\frac{1}{x+3}$ for $x < 1$, and $\sqrt{x - 1}$ for $x \geq 1$.

a. $(-\infty, -3) \cup (-3, \infty)$ ✓

b. $[1, \infty)$

c. All real numbers

d. $(-\infty, 1) \cup [1, \infty)$

For $x < 1$, $x + 3 \neq 0 \Rightarrow x \neq -3$. For $x \geq 1$, $x - 1 \geq 0$ is always true. Domain is all reals except $x = -3$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Piecewise Functions

Determine the range of $g(x)$ defined as $-x$ for $x < 0$, and $x + 1$ for $x \geq 0$.

a. $(0, \infty)$ ✓

b. $[1, \infty)$

c. $[0, \infty)$

d. All real numbers

For $x < 0$, $-x > 0$, giving outputs $(0, \infty)$. For $x \geq 0$, $x + 1 \geq 1$, giving $[1, \infty)$. Union is $(0, \infty)$.

30 [Intermediate] • Target: Domain and Range of Compositions

If $f(x) = \frac{1}{x-2}$ and $g(x) = \sqrt{x}$, find the domain of $(f \circ g)(x)$.

a. $[0, 4) \cup (4, \infty)$ ✓

b. $[0, \infty)$

c. $(4, \infty)$

d. All reals except $x = 2$

First, $g(x)$ requires $x \geq 0$. Second, $f(g(x)) = \frac{1}{\sqrt{x}-2}$ requires $\sqrt{x} \neq 2 \Rightarrow x \neq 4$. Domain: $[0, 4) \cup (4, \infty)$.

31 [Intermediate] • Target: Domain and Range of Compositions

If $f(x) = x^2$ and $g(x) = 3x - 1$, what is the range of $(f \circ g)(x)$ for $x \in \mathbb{R}$?

a. $[0, \infty)$ ✓

b. $[-1, \infty)$

c. $[1, \infty)$

d. All real numbers

$(f \circ g)(x) = (3x - 1)^2$. Any real squared is non-negative, so the range is $[0, \infty)$.

32 [Intermediate] • Target: Domain and Range of Compositions

Find the domain of $(g \circ f)(x)$ if $f(x) = 2x + 6$ and $g(x) = \ln(x)$.

a. $(-3, \infty)$ ✓

b. $[0, \infty)$

c. $(-6, \infty)$

d. $(3, \infty)$

$(g \circ f)(x) = \ln(2x + 6)$. The argument of logarithm must be strictly positive:
$2x + 6 > 0 \Rightarrow 2x > -6 \Rightarrow x > -3$.

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33 [Intermediate] • Target: Domain from equations

If $f(x) = \sqrt{x - 0}$ and $g(x) = x^2 - 1$, what is the domain of $f(g(x))$?

a. All real numbers

b. $x \leq -1$ **or** $x \geq 1$ ✓

c. $x \geq 1$

d. $-1 \leq x \leq 1$

Need $x^2 \geq 1$, so $|x| \geq 1$, giving $x \leq -1$ or $x \geq 1$.

34 [Intermediate] • Target: Domain and range from graphs

What is the domain and range of $y = \tan^{-1}(x)$? (variant 9)

a. **Domain** $(-\infty, \infty)$**, range** $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ ✓

b. Domain $(-\infty, \infty)$, range $(-\infty, \infty)$

c. Domain $(-\infty, \infty)$, range $[0, \pi]$

d. Domain $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, range $(-\infty, \infty)$

$\tan^{-1}(x)$ has domain $(-\infty, \infty)$ (restricted input) and range $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ (restricted principal output).

35 [Intermediate] • Target: Finding the range algebraically

What is the domain and range of $y = \sin^{-1}(x)$? (variant 5)

a. Domain $[-1, 1]$, range $(-\infty, \infty)$

b. Domain $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, range $[-1, 1]$

c. **Domain** $[-1, 1]$**, range** $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ ✓

d. Domain $(-\infty, \infty)$, range $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

$\sin^{-1}(x)$ has domain $[-1, 1]$ (restricted input) and range $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ (restricted principal output).

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36 [Intermediate] • Target: Domain and range in applied contexts

What is the domain and range of $y = \sin^{-1}(x)$? (variant 2)

a. Domain $[-\frac{\pi}{2}, \frac{\pi}{2}]$, range $[-1, 1]$

b. **Domain $[-1, 1]$, range $[-\frac{\pi}{2}, \frac{\pi}{2}]$** ✓

c. Domain $(-\infty, \infty)$, range $[-\frac{\pi}{2}, \frac{\pi}{2}]$

d. Domain $[-1, 1]$, range $(-\infty, \infty)$

$\sin^{-1}(x)$ has domain $[-1, 1]$ (restricted input) and range $[-\frac{\pi}{2}, \frac{\pi}{2}]$ (restricted principal output).

37 [Intermediate] • Target: Domain of composite and combined functions

What is the domain and range of $y = \sin^{-1}(x)$? (variant 10)

a. Domain $[-1, 1]$, range $(-\infty, \infty)$

b. Domain $[-\frac{\pi}{2}, \frac{\pi}{2}]$, range $[-1, 1]$

c. **Domain $[-1, 1]$, range $[-\frac{\pi}{2}, \frac{\pi}{2}]$** ✓

d. Domain $(-\infty, \infty)$, range $[-\frac{\pi}{2}, \frac{\pi}{2}]$

$\sin^{-1}(x)$ has domain $[-1, 1]$ (restricted input) and range $[-\frac{\pi}{2}, \frac{\pi}{2}]$ (restricted principal output).

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Domain from rational functions with factoring • Cross-Lesson: 7.01 Exponential Functions and Their Graphs

What is the domain and range of $y = \sin^{-1}(x)$? (variant 6)

a. Domain $[-1, 1]$, range $(-\infty, \infty)$

b. Domain $[-\frac{\pi}{2}, \frac{\pi}{2}]$, range $[-1, 1]$

c. **Domain $[-1, 1]$, range $[-\frac{\pi}{2}, \frac{\pi}{2}]$** ✓

d. Domain $(-\infty, \infty)$, range $[-\frac{\pi}{2}, \frac{\pi}{2}]$

$\sin^{-1}(x)$ has domain $[-1, 1]$ (restricted input) and range $[-\frac{\pi}{2}, \frac{\pi}{2}]$ (restricted principal output).

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39 [Challenging] • Target: Range using transformations • Cross-Lesson: 7.02 Logarithms and Their Properties

What is the domain and range of $y = \sin^{-1}(x)$? (variant 8)

a. **Domain** $[-1, 1]$**, range** $[-\frac{\pi}{2}, \frac{\pi}{2}]$ ✓

b. Domain $(-\infty, \infty)$, range $[-\frac{\pi}{2}, \frac{\pi}{2}]$

c. Domain $[-\frac{\pi}{2}, \frac{\pi}{2}]$, range $[-1, 1]$

d. Domain $[-1, 1]$, range $(-\infty, \infty)$

$\sin^{-1}(x)$ has domain $[-1, 1]$ (restricted input) and range $[-\frac{\pi}{2}, \frac{\pi}{2}]$ (restricted principal output).

40 [Challenging] • Target: Domain and range of piecewise functions • Cross-Lesson: 7.03 Exponential and Logarithmic Equations

What is the domain and range of $y = \sin^{-1}(x)$? (variant 7)

a. Domain $[-1, 1]$, range $(-\infty, \infty)$

b. Domain $(-\infty, \infty)$, range $[-\frac{\pi}{2}, \frac{\pi}{2}]$

c. Domain $[-\frac{\pi}{2}, \frac{\pi}{2}]$, range $[-1, 1]$

d. **Domain** $[-1, 1]$**, range** $[-\frac{\pi}{2}, \frac{\pi}{2}]$ ✓

$\sin^{-1}(x)$ has domain $[-1, 1]$ (restricted input) and range $[-\frac{\pi}{2}, \frac{\pi}{2}]$ (restricted principal output).

41 [Challenging] • Target: Finding Domains of Rational and Radical Functions • Cross-Lesson: 7.04 Applications: Growth and Decay

What is the domain of $f(x) = \sqrt{x - 4}$?

a. $x > 4$

b. $x \geq 4$ ✓

c. $x \geq -4$

d. $x \leq 4$

Applying Functions & Their Graphs principles, Need $x - 4 \geq 0$, so $x \geq 4$. Therefore, the verified correct option is $x \geq 4$.

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42 [Challenging] • Target: Finding Ranges of Algebraic Functions • Cross-Lesson: 8.01 Rational Functions and Asymptotes

What is the domain and range of $y = \cos^{-1}(x)$? (variant 5)

a. Domain $(-\infty, \infty)$, range $[0, \pi]$

b. **Domain $[-1, 1]$, range $[0, \pi]$** ✓

c. Domain $[-1, 1]$, range $(-\infty, \infty)$

d. Domain $[0, \pi]$, range $[-1, 1]$

$\cos^{-1}(x)$ has domain $[-1, 1]$ (restricted input) and range $[0, \pi]$ (restricted principal output).

43 [Challenging] • Target: Piecewise Functions • Cross-Lesson: 8.02 Graphing Rational Functions

Using the Remainder Theorem, find the remainder when $f(x) = 2x^2 - 1x + 6$ is divided by $x + 3$.

a. -27

b. 28

c. **27** ✓

d. 26

By the Remainder Theorem, the remainder equals $f(-3) = 2(-3)^2 - 1(-3) + 6 = 27$.

44 [Challenging] • Target: Domain and Range of Compositions • Cross-Lesson: 8.03 Introduction to Limits

What is the range of $f(x) = (x - 4)^2 + 3$?

a. $y \leq 3$

b. $y \geq 4$

c. All real numbers

d. $y \geq 3$ ✓

Since $(x - 4)^2 \geq 0$, the minimum value is $y = 3$, so range is $y \geq 3$.

45 [Challenging] • Target: Domain from equations • Cross-Lesson: 9.01 Arithmetic Sequences and Series

What is the domain of $f(x) = \frac{x + 1}{(x + 3)(x + 1)}$?

a. All real numbers

b. All reals except $x = -3$

c. **All reals except $x = -3$ and $x = -1$** ✓

d. All reals except $x = -4$

Denominator zero when $x = -3$ or $x = -1$; exclude both from the domain.

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